

SEORIM PARK

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South Korean

SKILLS:

- **Python, R:**
 - Data manipulation and visualization: **pandas, seaborn, matplotlib**
 - Generation of interactive plots using **plotly, dash, bokeh, Shiny**
 - Machine learning framework: **flax, tensorflow**
 - Pipeline development
- **C, C++, MATLAB:**
 - Implementation of mathematical models and algorithms
- **Git:**
 - Managed repositories and version control for large collaborative projects
- **GitHub link:**
 - <https://github.com/seorimpark>

LANGUAGE (CEFR):

- **Korean:** Mother tongue
- **English:** C1
- **French:** C2

PROFILE

As an engineer in Applied Mathematics, I am eager to apply my expertise in **the implementation and application of mathematical and statistical algorithms**, enabling **data-driven** approaches to **tackle real-life challenges**. I work **autonomously**, quickly grasp new concepts and workflows, and bring **enthusiasm** to collaborative environments supported by **effective communication skills**.

EDUCATION

École Polytechnique Fédérale de Lausanne (EPFL), *Lausanne, Switzerland*

- **Master of Science in Applied Mathematics**, Sept. 2021 - Feb. 2025
overall grade 5.15/6

École Polytechnique, *Palaiseau, France*

Sept. 2018 - July 2021

- **Bachelor of Science, graduated Cum Laude with a major in Mathematics & Computer Science and a minor in Biology**

WORK EXPERIENCE

Master's Thesis at the Chair of Dynamics of Learning Algorithms, EPFL

Sept. 2024 - Jan. 2025

- **Master's thesis in Deep Learning:** Explored mathematical foundations of recent deep learning methods such as Recurrent Neural Networks and Transformers, and conducted a comparative study on Linear Recurrent Units to observe the impact of hyperparameter tuning and model architecture on performance.

Internship at Nestlé Institute of Health Sciences (NIHS), Lausanne

June 2023 - Dec. 2023

- **Analyzed longitudinal multi-omics datasets** applying recent statistical techniques in **R** and **Python**.
- Developed **interactive data visualizations** in **Shiny** to **present key findings** to the team and department members.

PROJECTS

- Projects in **Statistics:** Collaborated on several projects involving statistical modeling and interactive visualization of real-world data and deployment of the application using plotly, dash in python and Shiny in R.
- Projects in **Machine learning:** Acquired advanced knowledge in Machine Learning and engaged in multiple projects, such as the comparative study of classifiers on label propagation for semi-supervised learning and the experimental study on the edge of stability of gradient descent in neural networks.
- Projects in **Mathematics** and **Computer Science:** Conducted a semester-long research project exploring network summary plots and implemented the method in python to test on real-world graph data. Worked on the implementation of Shortest Path algorithm in C on graph data with parallel computation.
- Projects in **database manipulation:** Collaborated on an interdisciplinary project to construct DNA sample database and API via MySQL.